

## **PPA 5**

### **Title**

### **Non-Projective Interfaces for Observing Substrate Computation**

### **Field of the Invention**

The invention relates to human-machine interfaces, interpretability, AI alignment, and computational observation systems.

### **Background**

All existing interfaces project internal system states into human-interpretable representations, collapsing spatial or nonlinear computation into symbolic forms.

This projection alters the system being observed.

### **Summary of the Invention**

The invention discloses methods for:

- Interacting with substrate computation without imposing objectives
- Observing invariants rather than representations
- Sampling persistence patterns without symbolic collapse

### **Detailed Description**

Observation mechanisms extract relational invariants, symmetry classes, or persistence statistics without mapping internal states to symbols or goals.

Interaction modifies constraints rather than issuing commands.

### **Key Novel Claims (Provisional)**

- Interfaces that do not impose representations
- Observation without semantic projection
- Interaction through constraint modulation
- Preservation of substrate autonomy